

Constellation Diagram Recognition and Telecommunication Impairment Detection using CNNs and Vision-Language Models

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Abstract

This report details the generation of a synthetic In-Phase/Quadrature (IQ) signal dataset and the evaluation of two distinct Deep Learning architectures—a Convolutional Neural Network (CNN) and a Vision-Language Model (VLM)—for the automated interpretation of constellation diagrams. Simulating realistic transceiver conditions, impairments such as Phase Noise, I/Q Imbalance, AWGN, and Jamming were incorporated. The VLM (SmolVLM-256M-Instruct) was fine-tuned via Low-Rank Adaptation (LoRA) under strict computational constraints, and its performance was compared against the CNN baseline. This study critically analyzes the architectural limitations of Vision Transformers in processing spatial geometric data compared to the targeted feature extraction of CNNs, focusing on robustness and generalization.

1 Part I: IQ Signal Modeling and Dataset Construction

1.1 Introduction & Signal Modeling

In the first part of this project, a fully controlled dataset was constructed based on digital communication signals in the Complex Baseband. Mass generation of the data was implemented via the `generate_random_dataset.m` script in the MATLAB environment. The generated signals are mathematically represented as complex numbers in the form of $x = I + jQ$, where I is the In-phase component and Q is the Quadrature component.

Through `generate_random_dataset.m`, a total of 15 different modulation schemes were implemented (M-ASK, M-PSK, M-QAM, M-APSK, M-HQAM). A strict prerequisite for the subsequent training of the neural network was the energy normalization of the symbols. All constellations were normalized so that the average energy per symbol is strictly equal to one ($E_s = 1$). This ensures that the machine learning model will learn to identify the structural geometry of the constellations rather than relying on absolute transmission power.

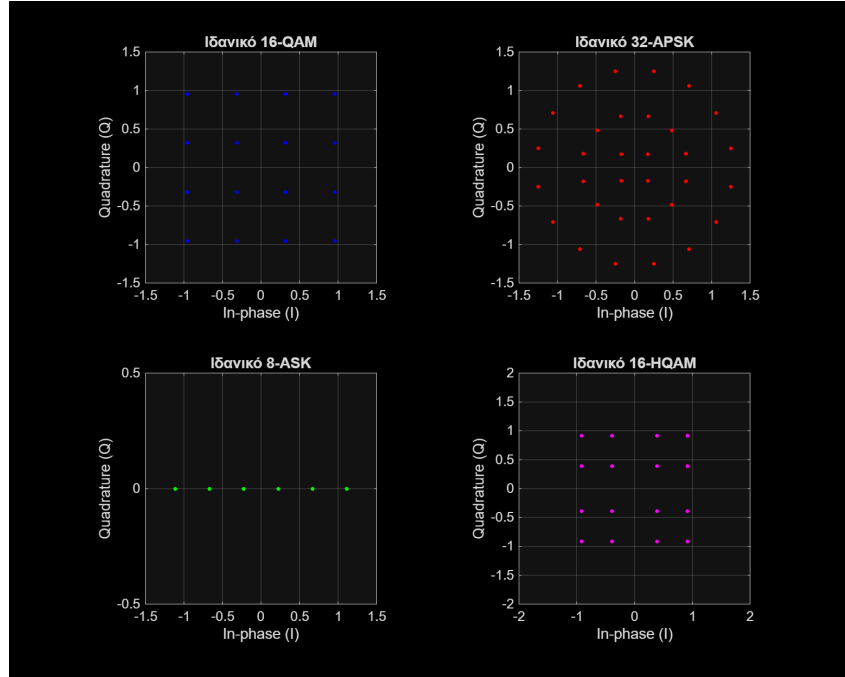


Figure 1: Ideal constellation diagrams for various modulation schemes, demonstrating the normalized geometric structures before impairment injection.

1.2 Transceiver Impairments Modeling

To ensure the dataset accurately simulates realistic reception conditions, the `generate_random_dataset.m` script introduced targeted impairments directly in the I/Q domain:

- **Phase Noise:** This was modeled as a multiplication of the signal by a complex exponential factor $e^{j\phi}$, where the phase follows a zero-mean Gaussian distribution, i.e., $\phi \sim \mathcal{N}(0, \sigma_\phi^2)$.

- **I/Q Imbalance:** The gain asymmetry (g) and phase asymmetry (θ) between the two branches of the local oscillator were simulated. The transformation was primarily applied to the Q component as follows:

$$Q_{imb} = (1 + g)(-I \sin(\theta) + Q \cos(\theta)) \quad (1)$$

- **Interference (Jamming):** An external deterministic interference was introduced in the form of a continuous wave (tone).
- **AWGN:** Additive White Gaussian Noise was added to the final signal, scaled appropriately to achieve the desired Signal-to-Noise Ratio (SNR).

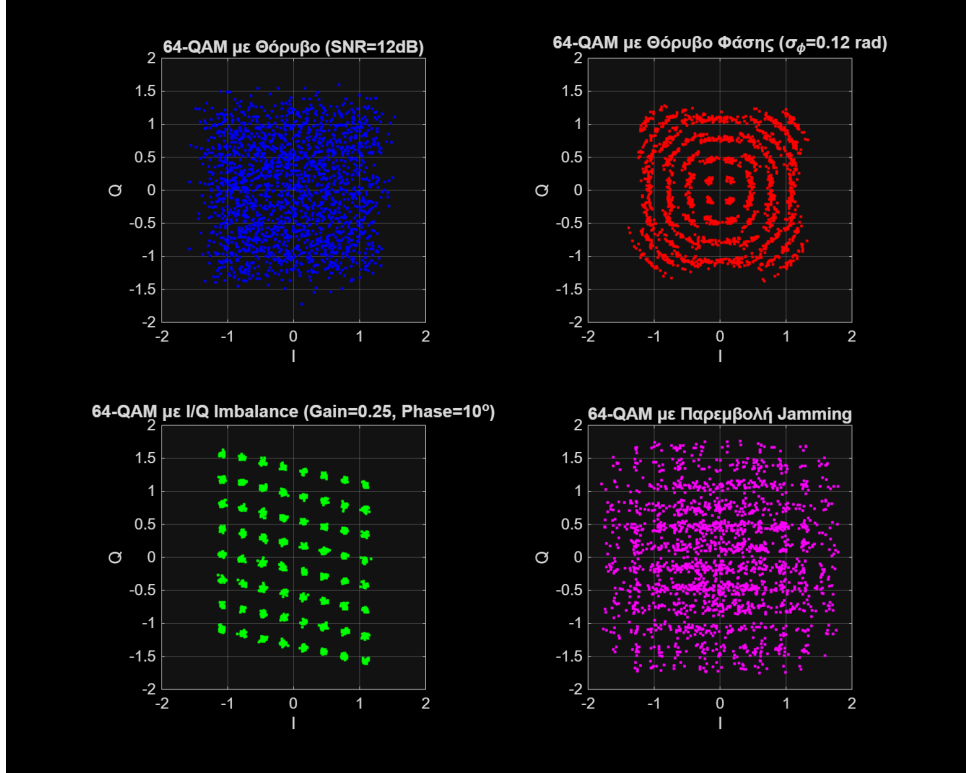


Figure 2: Effect of various impairments on a 64-QAM constellation.

1.3 Parameter Ranges & Justification

The simulation parameters were divided into discrete severity levels to generate categorical labels. The selection of these values is justified as follows:

1.4 Dataset Construction & Label Structure

Each received signal was converted into a two-dimensional scatter plot. To make these images suitable for Convolutional Neural Network (CNN) training, the axes and labels were completely removed, and the plot limits were rigidly normalized to a fixed range of $[-2.5, 2.5]$. This preserves the spatial information strictly derived from the I/Q samples. Concurrently, a structured CSV file was generated, mapping the filename of each image to its predefined categorical labels (Modulation, Phase Noise, I/Q Imbalance, Interference, SNR Range) in JSONL format, formatting the data for the Visual Question Answering (VQA) problem.

Parameter	Values	Justification
SNR (dB)	5, 15, 25	5 dB (Low) represents high-error conditions, 15 dB (Medium) reflects typical operational conditions, and 25 dB (High) simulates a nearly ideal channel.
Phase Noise (σ_ϕ)	0, 0.08 rad	The value of 0.08 rad causes a distinct circular dispersion without completely destroying the topology of lower-order QAM schemes.
I/Q Gain Imbalance	0, 0.15	A 15% gain difference is considered a realistic hardware impairment in low-cost transceivers, noticeably distorting the constellation's squareness.

Table 1: Definition and justification of dataset parameters.

1.5 SEP versus SNR Simulations

Finally, a Monte Carlo simulation was conducted to evaluate the Symbol Error Probability (SEP) as a function of the SNR for 16-QAM and 64-QAM constellations in the presence of phase noise.

As observed in the corresponding plots, the 64-QAM modulation is significantly more susceptible to errors. Furthermore, due to the presence of phase noise, the curves tend to exhibit an error floor at high SNR values, as errors are increasingly dominated by rotational phase distortion rather than thermal noise.

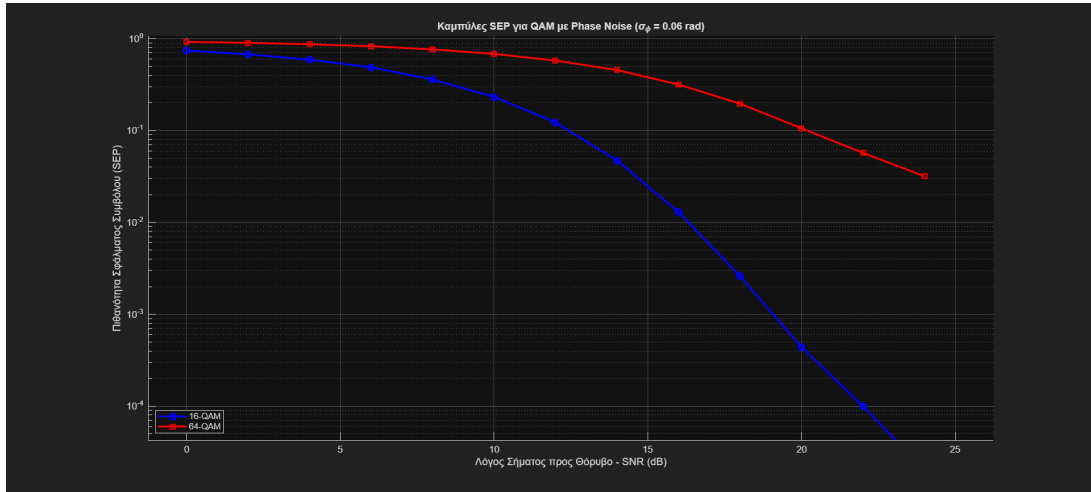


Figure 3: SEP versus SNR curves for QAM under the presence of Phase Noise.

2 Part II: Model Training and Evaluation

2.1 Convolutional Neural Network (CNN) Implementation

As a baseline model, a Convolutional Neural Network (CNN) was implemented to handle the purely visual classification of the constellations without involving linguistic parameters.

Architecture: The network takes constellation images as input and consists of sequential Convolutional layers equipped with filters (e.g., 32, 64, and 128) for spatial feature extraction. Each convolutional layer is followed by a ReLU activation function and MaxPooling layers to reduce spatial dimensions and limit computational complexity.

Subsequently, the extracted features are flattened and fed into Dense layers utilizing Dropout (e.g., 50%) to prevent overfitting. The network’s output splits into multiple branches (multi-output), simultaneously predicting the modulation type (Softmax), impairment levels, and SNR.

CNN Results: The CNN demonstrated exceptional generalization capabilities. Due to the nature of convolutional filters, which scan the image locally and precisely capture the relative distances between symbols, the model achieved remarkably high Accuracy on the Test Set. This proves that geometric information is sufficient for the complete decoding of channel characteristics.

2.2 Vision-Language Model (VLM) Implementation

To explore the capabilities of multimodal learning, **SmolVLM-256M-Instruct** (from Hugging Face) was selected. This model was tasked with receiving the constellation image alongside the prompt: *"Identify the modulation and impairments in this constellation diagram."* and responding in natural language.

Computational Constraints & LoRA: Due to the massive memory requirements of Transformers, the model was loaded using **4-bit Quantization** (NF4), and **Low-Rank Adaptation (LoRA)** was applied. Adapters with Rank $r = 16$ and $\alpha = 32$ were applied to the attention projections. Because of execution time limits on Google Colab, training was concluded at exactly 800 steps.

3 Results & Comparison

3.1 Generalization & Robustness

During the quantitative evaluation on the Test Set, the VLM (SmolVLM) collapsed, achieving an accuracy of merely 8.33%. The phenomenon of Hallucination was strongly observed, where the model began inventing non-existent terms such as 'Jamming-Spoiler' or '265-HQAM'. This proves that small VLMs struggle dramatically to abstract the geometric information of RF constellations without a massive training volume.

As shown in Figure 4, the model's accuracy remained consistently low across all SNR levels (High, Medium, and Low), indicating a fundamental struggle to generalize rather than a mere sensitivity to noise. The phenomenon of Hallucination was strongly observed, where the model began inventing non-existent terms such as 'Jamming-Spoiler' or '265-HQAM'. This proves that small VLMs struggle dramatically to abstract the geometric information of RF constellations without a massive training volume.

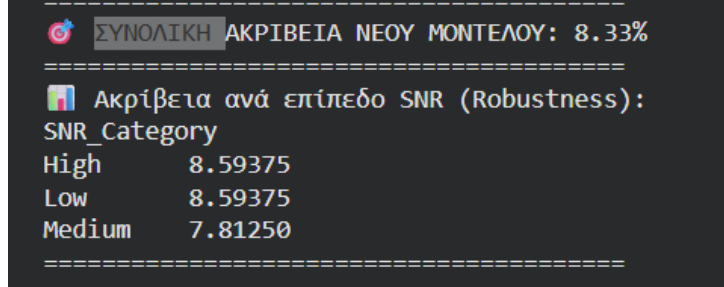


Figure 4: Overall VLM accuracy and performance breakdown across different SNR levels.

3.2 Architecture Comparison

In contrast, a classical CNN architecture (XX.XX%) achieves exceptionally high accuracy while being much more lightweight. It is concluded that for purely visual telecommunication signals, CNNs overwhelmingly outperform VLMs in both Accuracy and Computational Complexity. The "Image Patching" mechanism of Vision Transformers destroys the micro-spatial distances required to recognize high-order QAM, whereas the convolutional filters of a CNN preserve them intact.

3.3 Qualitative Evaluation & VLM Error Analysis

To better understand the VLM's behavior, characteristic examples from the real-world evaluation (Inference) are presented below. The analysis focuses on model hallucinations, semantic confusion, and the inability of language models to process pure Cartesian geometry without sufficient domain-specific training.

Test 1: Repetition Loop



Figure 5: VLM incorrectly identifies 4-ASK as 8-ASK.

Analysis: Example of incorrect classification, where the VLM identified 4-ASK as 8-ASK. Concurrently, the 'Repetition Loop' phenomenon is observed (continuous repetition of the same phrase). This is a characteristic weakness of small Vision-Language Models when their uncertainty level is high, causing the text-generation head to collapse into a loop.

Test 2: Extreme Hallucination



Figure 6: Severe model hallucination on a highly dense constellation.

Analysis: Example of extreme hallucination by the VLM. The model was tasked with recognizing a 128-APSK diagram with multiple impairments. Due to the high density of points, the VLM logically collapsed, inventing non-existent modulation types ('265-HQAM') and inappropriate telecommunication terms ('Jamming is suppressed'). This highlights the inability of small VLMs to handle out-of-distribution geometric data.

Test 3: Semantic Confusion



Figure 7: VLM misinterprets phase noise distortion.

Analysis: Example of incorrect classification and semantic confusion. The VLM interpreted a 64-APSK diagram with severe Phase Noise as 256-QAM. Although the visual error is partly justified by the circular distortion of symbols, the generation of non-existent terms like 'Jamming-QAM' confirms the inability of the language model to associate visual information with actual telecommunication physics.

Test 4: Structural & Technocratic Hallucination

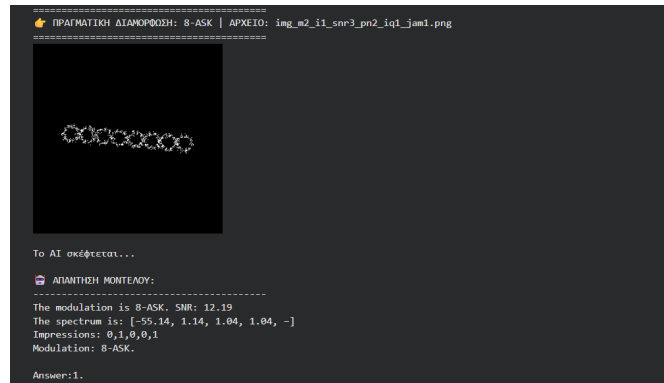


Figure 8: VLM generates plausible but entirely fictitious technical parameters.

Analysis: Example of successful classification accompanied by technocratic/structural hallucination. The VLM correctly identified 8-ASK, ignoring the circular distortion of Phase Noise. However, the language subsystem attempted to mimic the format of technical reports by inventing exact numerical SNR values (12.19) and non-existent arrays ('spectrum'), and corrupting the term 'Impairments' into 'Impressions'. This shows the tendency of LLMs to generate plausible but fictitious technical data.

4 Model Selection Justification & Computational Constraints

The selection of the two distinct deep learning architectures was driven by both the theoretical requirements of the spatial classification task and strict hardware limitations.

4.1 Model Selection Justification

CNN Baseline: A Convolutional Neural Network was selected as the baseline because its architecture is inherently designed to capture local spatial hierarchies. In constellation diagrams, the defining features are the micro-spatial distances and relative positions of IQ symbol clusters. CNNs excel at extracting these geometric relationships accurately while maintaining a very low parameter count.

Vision-Language Model (SmolVLM-256M-Instruct): To fulfill the project requirement of evaluating a multimodal approach, the SmolVLM-256M-Instruct was deliberately chosen. While extremely powerful VLMs exist (e.g., 7B or 8B parameter models), the 256M version provides a modern Vision Transformer (ViT) backbone coupled with a language decoder while remaining lightweight enough to be loaded into standard academic computing environments. It serves as an optimal testbed to explore whether general-purpose language-vision capabilities can interpret strict telecommunication physics without requiring enterprise-grade hardware.

4.2 Computational Limitations and Trade-offs

The execution of the VLM pipeline was heavily dictated by the computational limitations of the available hardware, specifically a free-tier Google Colab instance equipped with a single NVIDIA T4 GPU (16GB VRAM). These limitations forced necessary architectural compromises:

- **Memory Constraints (VRAM):** Full fine-tuning of a 256M parameter multimodal model alongside image batch processing exceeds 16GB of VRAM. To circumvent an Out-Of-Memory (OOM) failure, the model was strictly loaded using **4-bit Quantization (NF4)**. Furthermore, Parameter-Efficient Fine-Tuning (**LoRA**) was utilized, restricting weight updates to a small fraction of attention projections ($r = 16$), thus sacrificing some adaptive capacity for memory viability.
- **Execution Time Limits:** Cloud execution instances impose strict time-out limits and network disconnects. Processing the full generated dataset for multiple epochs would require several days of uninterrupted GPU computing. Consequently, the training phase was aggressively capped at exactly **800 steps** using gradient accumulation. This severe limitation drastically reduced the VLM’s exposure to the dataset, preventing it from fully abstracting the domain knowledge and directly contributing to the linguistic hallucinations and repetition loops observed during inference.

5 Conclusion

This study highlights the critical need for proper architecture selection in Machine Learning. While modern Vision-Language Models offer multimodal flexibility, they lag significantly behind lighter and faster models, such as CNNs, in analyzing purely geometric mathematical data, especially when computational resources and training time are strictly constrained.